MIL-F- 46481 (Ord)
27 January 1961

MILITARY SPECIFICATION
FUZE, MTSQ, M520 LESS BOOSTER

1. SCOPE

1.1 This specification covers one type of fuze, time and impact functioning, for use with high explosive, smoke and chemical ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATION

Military
- MIL-G-2550 - General Specification for Ammunition, Except Small Arms Ammunition

STANDARDS

Military
- MIL-STD-8 - Dimensioning and Tolerancing
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Inspection Terms and Definitions
- MIL-STD-350 - Jolt Test for use in Production of Fuzes
- MIL-STD-351 - Jumble Test for use in Production of Fuzes
- MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Ordnance Corps
- 30-1-7 - Standard for Dimensioning and Tolerancing
- D8594044 - Fuze, MTSQ, M520 Assembly
- IEL-8594044 - Inspection Equipment List for Fuze, MTSQ M520 Assembly

FSC 1390

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 No portion of this military specification shall be so construed as to permit variance or deviation from the functional requirements of the applicable drawings or specifications (see 6.6 and 6.7).

3.2 Preproduction Sample.- Unless otherwise directed by the contracting officer, a preproduction sample shall be required.

3.3 Waterproofness Test.- The fuze shall not display evidence of leakage when subjected to a vacuum which is two (2) inches of mercury less than the barometric pressure at time of test.

3.4 Jolt Test.- The fuze shall withstand the jolt when tested in accordance with Standard MIL-STD-350.

3.5 Jumble Test.- The fuze shall withstand the jumble when tested in accordance with Standard MIL-STD-351.

3.6 Drop Test.- No detonation or charring of the explosive elements shall occur when the fuze is subjected to a minimum drop of twenty-five (25) feet.

3.7 Ballistic Test.- The ballistic test shall consist of the following three (3) phases, and no premature burst shall occur (see 6.1 and 6.2).

a. Target Impact Phase
b. Safety Test Phase
c. Time Test Phase

3.7.1 Target Impact Phase.- The fuze shall not fail to cause high order detonation of the projectile when fired in accordance with 4.3.4.16.1.

3.7.2 Safety Test Phase.- There shall be no premature burst when fired in accordance with 4.3.4.16.2 (see 6.1 and 6.2).

3.7.3 Time Test Phase.- The fuze shall meet the requirement of Table I when tested in accordance with 4.3.4.16.3.

3.8 Workmanship.- Unless otherwise specified, workmanship shall be in accordance with Specification MIL-G-2550.

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TABLE I

Weapon	Elevation Degree Approx	Fuze Setting (Sec.)	Limits for Individual Time of Flight (Sec.) UCL	LCL	LIMITS FOR MEAN TIME OF FLIGHT Sample of 7 UCL	LCL	Sample of 6 UCL	LCL	Sample of 5 UCL	LCL	Sample of 4 UCL	LCL
155mm Howitzer Zone I	25	15	15.32	14.14	14.96	14.51	14.97	14.49	15.00	14.46	15.03	14.43
155mm Gun	17	30	29.34	27.42	28.74	28.02	28.77	27.99	28.81	27.95	28.86	27.90
155mm Gun	26	45	43.94	41.44	43.17	42.21	43.20	42.18	43.26	42.13	43.32	42.06
155mm Gun	34	60	58.58	55.70	57.69	56.59	57.72	56.56	57.79	56.49	57.86	56.42

Fuze Setting (Sec.)	LIMITS FOR MAX. DISPERSION Sample of 7 UCL	Sample of 6 UCL	Sample of 5 UCL	Sample of 4 UCL
15	1.03	1.01	.97	.93
30	1.66	1.62	1.57	1.49
45	2.18	2.12	2.05	1.96
60	2.50	2.44	2.37	2.26

UCL - upper control limit
LCL - lower control limit

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4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Provisions.- The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 to define the inspection terms used. The Ordnance Corps Standard for Dimensioning and Tolerancing 30-1-7, shall be used in conjunction with Standard MIL-STD-8 for drawing interpretation. Inspection shall be in accordance with Specification MIL-G-2550 except as specified herein.

4.1.1 Contractor Inspection.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor Quality Assurance System.- The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification and other applicable documents referenced in this specification. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.3 Government Verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

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4.2 Preproduction Sample Inspection.- Prior to the commencement of regular production, the contractor shall submit a preproduction sample to a Government approved facility for evaluation. The sample shall consist of 65 fuzes completely assembled and staked, 25 of each sub-assembly which shall be subjected to the static load and torque requirements, and 10 set of components part or assemblies as applicable for performance of the characteristic and tests listed in 4.3.3 and 4.3.4. The sample shall be manufactured in the same manner, using the same material, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the preproduction sample shall be inspected for all requirements of the drawings and specifications including all those characteristics listed in 4.3.3 and the tests in 4.3.4. The Government reserves the right to require new preproduction samples until such time as an acceptable sample is submitted.

4.3 Inspection Provisions.- Alternative inspection procedures and inspection equipment may be used by the contractor when procedures and equipment provide, as a minimum, quality assurance required in the contractual documents. The inspection procedures or equipment specified in this specification will prevail in case of dispute.

4.3.1 Material.- Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification of the material and part, the applicable requirements, the specific test performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction test sample. Prior to inspection, the inspector shall determine from the information stated on the test reports whether the material submitted complies with the requirements. Components made from material which does not comply shall be rejected.

4.3.2 Lot.

4.3.2.1 Submittal of Product.- The contractor shall submit the product in accordance with Standard MIL-STD-105 or Standard MIL-STD-414, as applicable.

4.3.3 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the criteria in Table II shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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TABLE II

Defect Classification

Major	AQL-0.25 percent
Minor	AQL-0.40 percent

4.3.3.1 Nose Assembly, Drawing 73-7-142E covering a detail of Drawing D8594044.

Defect	Method of Inspection
CRITICAL	
1. Distance from tip of firing pin to base of nose, min.	Gage
MAJOR	
101. Metal Defective	Visual
MINOR	
None	

4.3.3.2 Firing Arm Assembly, Drawing 73-7-225 covering a detail of Drawing D8594044.

CRITICAL	
1. Fracture of upright bend of arm	Visual
MAJOR	
None	
MINOR	
None	

4.3.3.3 Movement Assembly, (After Dynamic regulation) Drawing 73-7-137 covering a detail of Drawing D8594044.

CRITICAL	
1. Firing arm fires on not fire position	Gage
MAJOR	
101. Firing arm does not fire on must fire position	Gage
102. Firing pin protrusion less than min. in fired position	Gage
103. Firing arm shaft spring protrusion max.	Gage
104. Dynamic regulation stamp missing	Visual
105. Escapement lever not locked on proper side	Visual
106. Metal defective	Visual
107. Foreign matter	Visual
108. Burr	Visual

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Defect	Method of Inspection
MINOR	
201. Firing arm does not advance to "minimum travel"	Gage
202. Staking missing or inadequate	Visual

4.3.3.4 Movement Assembly, after assembly of timing disc assembly, Drawing 73-7-137 covering a detail of Drawing D8594044.

CRITICAL	
1. Timing disc bushing loose on knurl of the arbor (to be examined prior to assembly of locknut)	Manual
MAJOR	
101. Metal defective	Visual
MINOR	
None	

4.3.3.5 Movement Assembly, after assembly of timing disc retainer Drawing 73-7-137 covering a detail of Drawing D8594044.

CRITICAL	
1. Timing disc retainer insecure	Manual
MAJOR	
101. Timing disc safety disc location incorrect	Visual
102. Backlash excessive	Manual/Visual
103. Part missing	Visual
104. Improper assembly affecting functioning	Visual
105. Foreign matter	Visual
106. Burr	Visual
MINOR	
None	

4.3.3.6 Body, Movement and Lower Cap Assembly (Prior to Insertion of Sealing Plugs) Drawing 73-7-229 covering a detail of Drawing D8594044.

CRITICAL	
1. Safety disc off "Safe" position (to be checked after tensioning lock ring)	Visual
MAJOR	
101. Setting slot on lower cap and body misaligned	Gage
102. Setting pin extension below timing disc lug, min.	Gage
103. Improper assembly affecting functioning	Visual
104. Timing disc zero setting incorrect	Visual

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Defect	Method of Inspection
105. Part missing	Visual
106. Timing disc disengaged from setting pin	Visual
107. Setting pin loose	Manual
108. Body assembly screw missing or incorrect	Visual
109. Metal defective	Visual
110. Foreign matter	Visual
111. Burr	Visual

MINOR
None

4.3.3.7 Fuze Assembly (less booster) Drawing D8594044.

CRITICAL
None

MAJOR

101. Threads not full	Gage
102. Threads damaged	Visual
103. Assembly insecure	Manual
104. Marking incorrect, missing or illegible	Visual
105. Staking missing or inadequate (except sealing plugs)	Visual
106. Part missing	Visual
107. Point detonator closing disc punctured	Visual
108. Parts such as retainer or screws not flush or below mating surface	Visual
109. Waterproofing compound or varnish missing	Visual
110. Pull wire not inserted to proper depth	Visual
111. Waterproofing compound or varnish on lower cap and body joint	Visual
112. Metal defective	Visual
113. Foreign matter	Visual
114. Burr	Visual

MINOR

201. Pitch diameter of threads, max.	Gage
202. Pitch diameter of threads, min.	Gage
203. Major diameter of threads, min.	Gage

4.3.3.8 Sealed Fiber Container, Drawing (as specified in contract).

CRITICAL
None

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Defect	Method of Inspection
MAJOR	
101. Container cut or damaged through all unpregnated asphalt layers	Visual/Manual
102. Tape incomplete or badly wrinkled	Visual
103. Metal end loose or distorted	Visual
MINOR	
201. Gap between cover and body of container	Visual/Tactile
202. Contents loose	Manual
203. Marking incorrect, misleading or illegible	Visual
204. Tear - Tab - length improper	Scale/Trial
205. Color of tape incorrect	Visual

4.3.4 Inspection Testing.- The following tests shall be performed in accordance with provisions of 4.4 except as provided in 4.3.

4.3.4.1 Static Load and Assembly Torque.- The sample size and criteria for acceptance shall be as specified for range plans in Standard MIL-STD-414. An AQL of 0.25 percent shall be used. If the variable analysis indicates rejection but no individual value has actually failed to meet the drawing requirements, a comparable attributes plan shall be used, in which case the sample units used for the variable test shall be considered a part of the attributes sample.

4.3.4.1.1 Static Load.

Disc and Stud Assembly	Drawing 73-7-270B
Escapement Gear and Pinion Assembly	73-7-222A2
Escapement Lever Assembly	73-7-223A1
Firing Arm Assembly (Test of Pin and Arm)	73-7-225A2
No.1 Gear and Pinion Assembly	73-7-174A1
No.2 Gear and Pinion Assembly	73-7-175A2
No.3 Gear and Pinion Assembly	73-7-176A1
No.4 Gear and Pinion Assembly	73-7-177A1
No.5 Gear and Pinion Assembly	73-7-178A2

Above drawing covers a detail of Drawing D8594044.

4.3.4.1.2 Torque.

Escapement Gear and Pinion Assembly	Drawing 73-7-222A2
Disc and Stud Assembly	73-7-270B
Escapement Lever Assembly	73-7-223A1

Above drawing covers a detail of Drawing D8594044.

4.3.4.2 Torque of Timing Disc.- Each movement assembly shall be tested for compliance with the timing disc torque requirement of Drawing 73-7-137 covering a detail of Drawing D8594044. Failure of the assembly to conform with the requirement shall be cause for rejection of that assembly.

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4.3.4.3 Hardness.- Twenty samples from each lot of the following components shall be subjected to the hardness test. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot. The following drawings covers a detail of drawing D8594044.

Arm, Firing	Drawing 73-7-186A3
Guide, Adjusting	73-7-171B1
Lever, Escapement	73-7-224
Spring, Hammer	73-7-231
Spring, Timing Disc	73-7-103A4
Washer, Timing Disc	73-7-103C4

4.3.4.4 Firing Arm Bend.- The same 20 firing arms that complied with the hardness requirement shall be subjected to the bend test. Failure of any firing arm to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot.

4.3.4.5 Timing Disc Spring Dead Load.- The same 20 springs that complied with the hardness test shall be subjected to the dead load test. Failure of any spring to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot.

4.3.4.6 Compression.- Twenty samples from each lot of the following components shall be subjected to the compression test. Failure of any spring to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

Component	Drawing
Firing Pin Spring	73-7-103 covering a detail of Drawing D8594044

4.3.4.7 Dynamic Regulation.- Each movement assembly shall be dynamically regulated. Failure of any assembly to comply with the requirement specified on the applicable drawing shall be cause for rejection, subject to correction and resubmission.

4.3.4.8 Non-Arming.- Each point detonator assembly shall be subjected to a non-arming test. Failure of any assembly to comply with the requirement of the applicable drawing shall be cause for rejection of the assembly.

4.3.4.9 Arming.- Each point detonator assembly shall be subjected to an arming test. Failure of any assembly to comply with the requirement of the applicable drawing shall be cause for rejection of the assembly.

4.3.4.10 Setting Torque.- (Body and Lower Cap).- Each assembly shall be tested to determine compliance with the requirement of the applicable drawing. Any assembly failing to conform with the requirement shall be rejected.

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4.3.4.11 Safety Lever Assembly (Arming).- (Movement Sub-Assembly).- Each movement sub-assembly shall be tested to determine compliance with the arming requirement of the applicable drawing. Any assembly failing to conform with the requirement shall be rejected.

4.3.4.12 Safety Lever Assembly (Non-Arming) (Movement Sub-Assembly).- Each movement sub-assembly shall be tested to determine compliance with the Non-Arming requirements of the applicable drawing. Any assembly failing to conform with the requirement shall be rejected.

4.3.4.13 Hammer Spring Deflection.- Each housing and hammer spring assembly shall be tested to determine compliance with the requirement specified on the applicable drawing. Any assembly failing to conform to the requirement shall be rejected.

4.3.4.14 Kinetic Energy of Firing Pin.- Each movement assembly shall be tested to determine compliance with the requirement of the applicable drawing. Any assembly failing to conform to the requirement shall be rejected.

4.3.4.15 Production Lot.

4.3.4.15.1 Shop Tests.- Twenty (20) complete fuze assemblies shall be selected from each lot and subjected to the following tests.

4.3.4.15.2 Waterproofness Tests.- Ten (10) fuzes shall be subjected to the waterproofness tests. The lot shall be accepted provided not more than one (1) fuze fails to meet the requirements of 3.4.

4.3.4.15.3 Jolt Test.- Five (5) fuzes which have passed the waterproofness test shall be subjected to the jolt test. Failure to meet the requirement of 3.5 shall be cause for rejection of the lot.

4.3.4.15.4 Jumble Test.- Four (4) fuzes which have passed the waterproofness test shall be subjected to the jumble test. Failure to meet the requirement of 3.6 shall be cause for rejection of the lot.

4.3.4.15.5 Twenty-five (25) Foot Drop Test.- Ten (10) fuzes shall be subjected to the drop test. Failure to meet the requirement of 3.6 shall be cause for rejection of the lot.

4.3.4.16 Ballistic Tests.- The Government will perform ballistic tests on 45 fuzes from each lot as follows:

4.3.4.16.1 Target Impact Phase.- Ten (10) fuzes set on safe (S) shall be fired from a 75mm gun, against a one (1) inch plywood bursting screen. The lot shall be rejected if more than one (1) fuze fails to meet the requirement of 3.7.1.

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4.3.4.16.2 Safety Test Phase.- Seven (7) fuzes shall be set for ten (10) seconds and fired in a 76mm gun. Failure to meet the requirement of 3.7.2 shall be cause for rejection of the lot.

4.3.4.16.3 Time Test Phase.- One (1) group of seven (7) fuzes shall be fired in a 155mm Howitzer and three (3) groups of seven (7) fuzes shall be fired in a 155mm gun. The lot shall be accepted if not more than three (3) time failures, or three (3) off times, or two (2) time failures and one (1) off time, or two (2) off times and one (1) time failure, shall occur in the time test phase. Individual time of flight, mean times of flight and maximum dispersions shall be in accordance with the applicable limits contained in TableI (see 6.3 and 6.4). Off times will be disregarded in computation of the mean and the maximum dispersion. The sample number will be equal to the number of values of times of flight used (excluding time failures and off times) in calculating the mean. Duds shall be discounted entirely and shall not be considered as time failures (see 6.5).

4.3.5 Inspection Equipment.- The examination and tests shall be made using equipment listed on IEL-8594044. The Government will determine by means of its monitoring program that the contractor has available and correctly utilizes gaging, measuring and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test Methods and Procedures.

4.4.1 Static Load.- Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. All units shall be tested to the point of component disassembly. The values obtained shall be used to determine lot acceptability in accordance with the specified variable plan. All samples tested shall be scrapped.

4.4.2 Torque.- Escapement lever assemblies shall be tested prior to insertion of the escapement spring. All units shall be tested to the breakaway point. The values obtained shall be used to determine lot acceptability in accordance with the specified variables plan. All samples tested shall be scrapped.

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4.4.3 Timing Disc Torque.- The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the locknut on the arbor. The highest and lowest reading will be taken in one complete revolution of the timing disc for determination of compliance with the requirements.

4.4.4 Hardness.- Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The average value of 3 hardness reading taken on each sample unit shall be used as the individual value.

4.4.5 Firing Arm Bend.- The test shall be performed as specified on the applicable drawing. The firing arms that have been subjected to the bend test shall be scrapped.

4.4.6 Timing Disc Dead Load.- The test shall be performed using the values specified on the applicable drawing. Timing Disc springs that have been subjected to the dead load test shall be scrapped.

4.4.7 Compression.- The firing pin springs shall be compressed for the length of time specified on the applicable drawing. Springs that have been subjected to the compression test shall be scrapped.

4.4.8 Dynamic Regulation.- The movement assembly shall be regulated using Government approved equipment, the operational procedures furnished with the equipment shall be used. The spinning chuck shall rotate at a constant speed of 12,600 RPM $\pm$ 55 RPM. Timekeeping and performance of the movement shall be checked over a period of not less than forty (40) seconds. No adjustment shall be made to the escapement during the (40) second check run. Check runs shall be started with the drive gears in fully wound position. Should any movement require more than (4) check runs, it shall be rejected, subject to the insertion of a new escapement spring. The old escapement spring shall be scrapped. The technical agency shall be informed of the regulating frequency being used.

4.4.9 Non-Arming (Point Detonator, Prior to Loading).- The assemblies shall be spun at the non-arming RPM specified on the applicable drawing. Non-conforming assemblies may be reworked and resubmitted.

4.4.10 Arming (Point Detonator, Prior to Loading).- The assemblies shall be spun at the arming RPM specified on the applicable drawing. Non-conforming assemblies may be reworked and resubmitted.

4.4.11 Setting Torque.-(Body and Lower Cap).- The assemblies shall be placed in the fixture and the lock ring tightened until the torque necessary to turn the movement housing meets the drawing requirement. The set screws are then to be tightened to secure retaining ring. Non-conforming assemblies may be reworked and resubmitted.

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4.4.12 Safety Levers Assembly (Movement Sub-Assembly).- The non-arming test shall be performed on completely wound sub-assemblies with the backlash removed. The sub-assemblies shall be rotated at the RPM specified on the applicable drawing, for a time interval sufficient to determine compliance with the requirement.

4.4.13 Safety Lever Assembly (Movement Sub-Assembly).- The arming test shall be performed on completely wound sub-assemblies with the backlash removed. The sub-assemblies shall be rotated at the RPM specified on the applicable drawing for a time interval sufficient to determine compliance with the requirement.

4.4.14 Hammerspring Deflection (Housing and Hammer Spring Assembly).- The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on the applicable drawing. The hammerspring must return to surface "C" when released. Non-conforming assemblies may be reworked and resubmitted.

4.4.15 Kinetic Energy of Firing Pin (Movement Sub-Assembly).- The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position. Non-conforming assemblies may be reworked by insertion of a new firing pin spring and resubmitted.

4.4.16 Waterproofness Test.- The fuze shall be completely submerged in water ($70^{\circ}F \pm 15^{\circ}F$) and subjected to the required vacuum for a minimum of twenty (20) seconds.

4.4.17 Jolt Test.- Fuzes which have been subjected to the waterproofness test shall be selected, and tested in accordance with MIL-STD-350. After breakdown and examination, all components shall be scrapped.

4.4.18 Jumble Test.- Fuzes which have been subjected to the waterproofness test shall be selected, and tested in accordance with MIL-STD-351. After breakdown and examination, all components shall be scrapped.

4.4.19 Twenty-five (25) Foot Drop Test.- The fuzes shall be set at intervals of 7.5 seconds, with pull wire removed, and shall be assembled to a three (3) inch AA Shell, M42 fitted with an inert M20 booster or equivalent adapter. The shell shall be filled with inert material to give a service weight of 12 3/4 pounds + 1/4 pound. It shall be dropped, base downward, from a height of twenty-five (25) feet in such a manner as to fall freely and impact upon a plate 3/4 $\pm$ 1/32 inch thick. The plate shall be made of lead antimony alloy containing 10% maximum by weight of antimony, cast in a confining ring of steel to prevent the free flow of the alloy under impact. The lead antimony plate shall be placed on a steel anvil at least two (2) inches thick.

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4.4.20 Ballistic Tests.- The following tests will be performed by the Government.

4.4.20.1 Target Impact Phase.- The fuzes shall be assembled with M125 Boosters, set on safe (S) and fired against a one (1) inch plywood bursting screen placed not less than 400 feet from the muzzle of the gun. A 75mm gun employing a 112% normal pressure shall be used.

4.4.20.2 Safety Test Phase.- The fuzes shall be assembled with M125 Boosters, and fired in a 76mm gun, M1A1 or M1A2 at approximately 2,700 feet per second muzzle velocity. The fuzes shall be set for ten (10) seconds. The timing results of the fuzes of this group will not be used as a basis for acceptance or rejection of the lot, except that a premature burst shall be cause for rejection.

4.4.20.3 Time Test Phase.- One (1) group of fuzes shall be assembled with M125 boosters and fired in a 155mm Howitzer, Zone 1, and three (3) groups of fuzes each assembled with M125 booster shall be fired in a 155mm gun at approximately 2800 feet per second muzzle velocity. All settings shall be made using a hand wrench, utilizing the graduations on the body for accurate setting. For computing mean time of flight and maximum dispersion use only fuze chronograph time or corrected clock time if the time to burst for any round is lost by the fuze chronograph. The correction factor for corrected clock time shall be the mean difference between the electric clock time and the fuze chronograph time of each of the other fuzes in the same group. Subtract the mean difference time from the electric clock time of the fuze time that was lost to determine its chronograph time. Disregard any three (3) of the allowable time failures or off times in computation of mean time of flight and maximum dispersion.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging and Marking.- Preservation and packaging shall be level "A". Marking shall be in accordance with instruction issued by the contracting officer.

5.2 Data Card.- Data card information shall be as specified in specification MIL-G-2550.

6. NOTES

6.1 A premature burst is considered to have occurred in the Time Test Phase or the Safety Test Phase if the round detonates in the bore of the weapon or if the time of flight is less than 1.5 second.

6.2 A premature burst in the Target Impact Phase is defined as a round detonation in the bore of the gun or in flight before impact.

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6.3 An off time is any individual air burst time occurring outside the limits established for individual times of flight in Table I.

6.4 A time failure in the time phase is considered to have occurred if the round fails to detonate during flight and detonation occurs upon subsequent impact.

6.5 A dud in the time phase is defined as a round which fails to detonate during flight and on subsequent impact.

6.6 All components should be manufactured in strict accordance with Ordnance drawings but the dimensions and tolerances specified represent only one set of conditions which will result in satisfactory performance. Numerous other combinations of dimensions and tolerances may also prove satisfactory. Therefore, it is not mandatory that dimensional requirements on Ordnance Drawings be complied with, provided that departure from specified requirements do not materially affect the basic design of the fuze. This relaxation of requirements applies to dimensions only and shall not apply to those dimensions which effect the external profile of the fuze or those otherwise qualified in this specification. The foregoing is not to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.7 The contractor shall maintain records showing intentional deviations from dimensional requirements, the extent of such deviations, the amount of material involved and the identity of the lots into which the material is assembled.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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